Aland Edbert Manurung

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EDUCATION

University of Leicester **Leicester, United Kingdom**
Satellite Data Science/M.Sc. *September 2021 – 2022 (ongoing)*

- Chairman of Indonesian Student Association in Leicestershire (PPI Leicestershire), and Department Reps of School of Geology, Geography, and the Environment.
- Relevant modules: Fundamentals of GIS, R for Data Science, Big Data and Predictive Analytics, Geospatial Database and Information Retrieval, Satellite Data Analysis in Python, Earth Observation of Atmosphere.

Institut Teknologi Bandung **Bandung, Indonesia**
Geological Engineering/S.T. (B.Sc.) *October 2020*

- GPA 3.44, 2nd winner of English Debate on Energy Security by AAPG Unpad.
- Relevant modules: Math and Statistics for Geology, Geostatistics, Environmental Geology.

WORKING EXPERIENCE

Kita Earth Limited **London, United Kingdom**
Above Ground Biomass Analyst, Part-Time *May 2022 – September 2022*

- Tasked to develop above ground biomass predictor using optical and SAR (if applicable) remote sensing data.
- Tasked to do carbon accounting using existing carbon model.

Center for Landscape and Climate Research, University of Leicester **Leicester, United Kingdom**
Research Assistant, Part-Time *Jan 2022 – March 2022*

- Developed machine learning model to predict biomass from LiDAR dataset. Communicate workflows of data acquisition, took field samples of biomass, and analyze LiDAR parameters.

PT Bhumi Varta Technology **Tangerang, Indonesia**
GIS Developer Intern, Internship *May 2021 – July 2021*

- Developed MOWL, mobile app of indoor GIS for shopping mall and Developed machine learning models for Land-use Classification based on satellite imagery.

CERTIFICATION AND QUALIFICATION

- BCG – Data Science and Advanced Data Analytics Virtual Experience Program *February 2022*
- BCG – Climate and Sustainability Virtual Experience Program *February 2022*
- Geo University – Overview of LiDAR; system variations, data interpretation & applications *January 2021*
- Udemy – Python for Data Science and Machine Learning Bootcamp *July 2021*
- Udemy – Geospatial Data Science with Python: Geopandas *July 2021*
- IHE Delft – Programming for Geospatial Hydrological Applications *April 2021*

SKILLS

Geographic Information System (QGIS, ArcGIS, Geopandas – Python, Terra – R), Data science (Python, R, PostgreSQL), Office operation software (Microsoft Office – Word, Excel, Powerpoint, Publisher), Media and publishing software (Scribus, GIMP, Inkscape), Video production (Hitfilm)

COURSEWORK PROJECTS

GY7701: Fundamentals of GIS **University of Leicester, Leicester**

Analysis of Bank Branch Closure and New Branch Potential at Isle of Wight using Network Analysis and Artificial Neural Network

- Done network analysis to generate bank service areas in 2019 for walking and cycling distance.
- Developed artificial neural network to identify new bank branch potential based on age structure, economic activity, occupation, and car ownership of residents at Isle of Wight.

GY7702: R for Data Science **University of Leicester, Leicester**

Geodemographic Analysis of Havering to Support Local Development

- Analyzed health, education, and occupation variables correlation and done dimensionality reduction using Principal Component Analysis (PCA) method.
- Built K-means model for clustering demographic classifications based on health, education, and occupation characteristic of groups of people in Havering.

GY7705: Remote Sensing **University of Leicester, Leicester**

Assessing Indonesia Peat Degradation caused by Peat Fire in 2015 using Multiple Remote Sensing Data

- Analyzed peat degradation using optical and SAR. Analyzed burn scar using SICI (Shortwave infrared combustion index) and surface deformation using Interferometry SAR technique.

GY7708: Geospatial Database and Information Retrieval **University of Leicester, Leicester**

Geographic Text Analysis of Selton from Wikipedia Pages

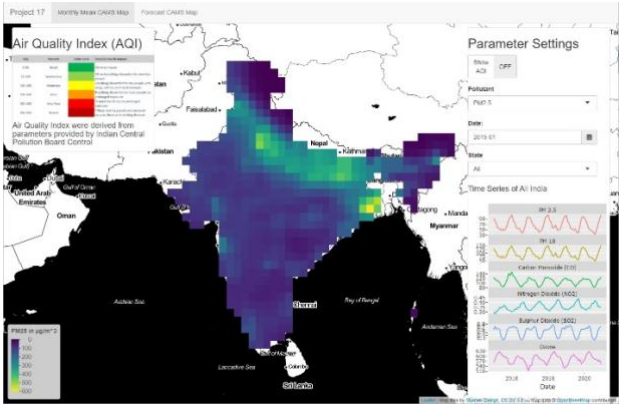
- Retrieved summary geocoded pages from Wikipedia and analyzed text using natural processing language (NLP) techniques.
- Generated map of sentiments in Selton based on Wikipedia pages.

GY7709: Satellite Data Analysis in Python **University of Leicester, Leicester**

Analysis of Drought at Terai Regions in Uttarakhand using NDWI

- Retrieved Sentinel-2 images between July and December 2021 using Python and generated time-series analysis of NDWI (Normalized difference water index)
- Created interactive map to visualize time-series of NDWI in video format.

PORTFOLIO

[UAMS – Urban Air Monitoring System]

UAMS is a webapp to display PM2.5, CO, NO_2, and SO_2 on defined month and year. Data were derived from CDS API from Copernicus Atmosphere Monitoring Service and integrated with R Shiny. This project is a group coursework project of Earth Observation of Atmosphere module in University of Leicester, I worked as the developer of the webapp.